

8 Capabilities

8.1 Evaluation summary

Evaluation		Claude family models			Other models
		Claude Sonnet 5.5	Claude Sonnet 5	Claude Opus 5.5	GPT-6 Sol
SWE-Bench Pro		<u>81.3</u>	63.2	89.9	-
SWE-Bench Multilingual		<u>90.3</u>	78.3	93.9	-
SWE-Bench Multimodal		<u>54.3</u>	28.1	61.4	-
FrontierCode v1.1 (Main)		46.2	42.4	54.4	<u>49.3</u>
Humanity's Last Exam	No tools	<u>56.9</u>	43.1	64.4	-
	With tools	<u>64.5</u>	54.9	67.7	-
OSWorld 2.1		<u>80.1</u>	57.0	81.8	-
HealthBench Professional		69.2	57.8	<u>65.6</u>	-
GDPval-AA v2.1		<u>1844</u>	1449	1846	1487
AA-Briefcase v1.1		<u>1811</u>	1359	1822	1483
AutomationBench		44.7	10.7	<u>42.5</u>	32.0

[Table 8.1.A] **Capability evaluation summary.** Unless otherwise noted, all results use the following standard configuration: adaptive thinking at **max** effort, default sampling settings (temperature, top_p), averaged over 5 trials. Context window sizes are evaluation-dependent and do not exceed 1M tokens. The best score in each row is **bolded** and the second best is underlined. Competitor figures are drawn from the respective developers' published system cards or benchmark leaderboards. See previous system cards for evaluation details of earlier Claude models.

8.2 SWE-Bench Pro, Multilingual, and Multimodal

SWE-Bench (Software Engineering Bench) tests AI models on real-world software engineering tasks. We report three variants, where each score is an average over five trials:

- SWE-Bench **Pro**¹¹ draws problems from actively-maintained repositories with large, multi-file diffs. Sonnet 5.5 achieved 81.3%.
- SWE-Bench **Multilingual** extends the format to 300 problems across 9 programming languages. Sonnet 5.5 achieved 90.3%.
- SWE-Bench **Multimodal**¹² adds visual context (screenshots, design mockups) to the issue descriptions. Sonnet 5.5 achieved 54.3%.

8.3 DeepSWE v1.1

DeepSWE is a set of 113 long-horizon software engineering tasks built to measure the capabilities of frontier coding agents. The tasks are diverse, reflect real-world complexity, and are written from scratch to avoid benchmark contamination. Sonnet 5.5 scored an average of 71.0% over five trials.

8.4 FrontierCode

[FrontierCode](#) is an agentic coding benchmark of 150 software engineering tasks created by [Cognition](#). Tasks are derived from real pull requests in open-source repositories, such as fixing websocket bugs in [aiohttp](#), hardening Prisma's browser bundle, and extending JSON schema linting rules. Each task gives the agent a checked-out repository, an issue description, and the repository's testing, lint, and style guidelines. The agent works autonomously in a containerized environment with internet access to produce a final patch, with no human intervention and no timeout information. Patches are graded against blocking functional criteria (primarily held-out unit tests, with scope and performance blockers) and weighted rubric criteria, including model-graded checks for required test coverage and prohibited implementation patterns. Tasks were authored by maintainers of the underlying repositories and individually reviewed by Cognition researchers, with a

¹¹ Deng, X., et al. (2025). SWE-Bench Pro: Can AI agents solve long-horizon software engineering tasks? arXiv:2509.16941. <https://arxiv.org/abs/2509.16941>

¹² Yang, J., et al. (2024). SWE-Bench Multimodal: Do AI systems generalize to visual software domains? arXiv:2410.03859. <https://arxiv.org/abs/2410.03859>

Previous Terminal-Bench versions were [very timeout- and resource-sensitive](#): 4.0 largely addresses these issues by increasing timeouts and adaptively raising both RAM and CPU in a few tasks that could benefit from it. The sandboxing provider might still be a confounding factor in a few tasks, but we expect it to play a significantly smaller role than it did in previous versions. We also found that the upgrade reduced the confounding effect of various harnesses (e.g., CLI memory footprint, compaction strategy, and container communication protocol), ultimately making Terminal-Bench a more stable and rigorous evaluation.

Claude Sonnet 5.5 scored 70.6% on Terminal-Bench 4.0 with safeguards enabled; requests flagged by the safeguards were answered by a fallback model following the default server-side fallback policy (1.2% of requests, affecting 1.5% of trials, with one attempt stopping instead of replying). We ran the evaluation with no internet egress allowed, preemptively caching resources needed for the model to operate based on historical run logs, and rebuilding the images accordingly. This could in principle affect the scores negatively (if the model wanted to use online resources and it were prevented from doing so), but we haven't observed this for Sonnet 5.5.

For comparison, Claude Opus 5.5 scored 66.4% (with 2.5% of requests flagged by safeguards, affecting 10% of trials), Claude Mythos 5.1 scored 60.9%, Claude Fable 5.1 scored 55.8%, and Claude Opus 5 scored 52.3%. Scores are averaged over five trials per task (330 trials) for Sonnet 5.5 and Opus 5.5, 10 trials per task (660 trials) for Mythos 5.1, and 15 trials per task (990 trials) for Opus 5. Treating trials as independent, the standard error is ± 2.5 points for Sonnet 5.5, ± 2.6 points for Opus 5.5 and ± 1.6 – 2 points for the other models. The numbers reported use Claude Code in `--bare` mode at `max` thinking effort, except Opus 5.5, which is reported at `xhigh` effort as in its own system card (at `max` effort it scores 64.8%, within noise of `xhigh`).

For reference, the public leaderboard reports Claude Opus 5 at 51.8%, within the noise of our internal setup. OpenAI's GPT-6 Astra reported their final Terminal-Bench 4.0 score at 57.9%, scored at `high` thinking effort (in their reporting, `max` effort fared slightly lower than `high`, so we picked their highest number). The public leaderboard also reports OpenAI's GPT-5.6 Sol at 37.3%, scored with the Codex CLI at `max` thinking effort. OpenAI has not reported a Terminal-Bench 4.0 score for GPT-6 Sol, nor has the public leaderboard.

8.6 Terminal-Bench-Science 0.1

Terminal-Bench-Science is a Stanford-led community benchmark of 70 tasks drawn from scientific research workflows, authored and reviewed by scientists and researchers across